HYEON JEON

Ph.D Student, Dept. of Computer Science and Engineering @ SNU
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EDUCATION

Ph.D Student | ***Dept. of Computer Science and Engineering*** Sep.2022 – Present
Seoul National University Seoul, Republic of Korea

- Human-Computer Interaction Laboratory (advised by Dr. Jinwook Seo)

MS Student | ***Dept. of Computer Science and Engineering*** Sep.2020 – Aug.2022
Seoul National University Seoul, Republic of Korea

- Switched to Ph.D program in Sep. 2022

Bachelor of Science in Computer Science & Engineering Mar.2016 – Aug.2020
POSTECH Pohang, Republic of Korea

- Early graduation (seven semesters)
- Graduated with honors (Magna Cum Laude)

RESEARCH INTEREST

Visual Analytics, High-dimensional & Large-scale Data Analysis
Clustering, Visual & Computational Cluster Analysis

PUBLICATIONS

PREPRINTS

[P02] H. Jeon, M. Aupetit, D. Shin, A. Cho, S. Park, and J. Seo, "Sanity check for external clustering validation benchmarks using internal validation measures," *arXiv:2209.10042*

[P01] H. Jeon, M. Aupetit, S. Lee, H.-K. Ko, Y. Kim, and J. Seo, "Distortion-aware brushing for interactive cluster analysis in multidimensional projections," *arXiv:2201.06379*

INTERNATIONAL JOURNALS

[J02] H. Jeon, H.-K. Ko, J. Jo, Y. Kim, and J. Seo, "Measuring and explaining the inter-cluster reliability of multidimensional projections," *IEEE Transactions on Visualization and Computer Graphics (TVCG, Proc. VIS)*, vol. 28, no. 1, pp. 551–561, 2022

[J01] Y. Kim, J. Kim, H. Jeon, Y.-H. Kim, H. Song, B. Kim, and J. Seo, "Githru: Visual analytics for understanding software development history through git metadata analysis," *IEEE Transactions on Visualization and Computer Graphics (TVCG, Proc. VAST)*, vol. 27, no. 2, pp. 656–666, 2021

PEER-REVIEWED CONFERENCES

[C04] H.-K. Ko, G. Park, H. Jeon, J. Jo, J. Kim, and J. Seo, "Large-scale text-to-image generation models for visual artists' creative works," in *Proceedings of the 28th ACM International Conference on Intelligent User Interfaces (IUI)*, 2023. to appear.

[C03] H. Jeon*, H.-K. Ko*, S. Lee, J. Jo, and J. Seo, "Uniform manifold approximation with two-phase optimization," in *2022 IEEE Visualization and Visual Analytics (VIS)*, pp. 80–84, 2022. (**equal contribution.*)

[C02] S. Park, S. Lee, Y. Kim, H. Jeon, S. Jung, J. Bok, and J. Seo, "Vant: A visual analytics system for refining parallel corpora in neural machine translation," in *15th IEEE Pacific Visualization Symposium (PacificVis)*, 2022. to appear. received ***Best Visualization Notes (Top*** 1st ***Submission)***

[C01] Y. Kim, H. Jeon, Y.-H. Kim, Y. Ki, H. Song, and J. Seo, "Visualization support for multi-criteria decision making in software issue propagation," in *14th IEEE Pacific Visualization Symposium (PacificVis)*, pp. 81–85, 2021

Posters and Domestics

[O02] S. Jung, D. Shin, J. Bok, S. Park, H. Jeon, J. Seo, I. Lee, and S. Park, "Interactive visual analytics system for criminal intelligence analysts with multiple coordinated views," *Journal of KIISE*, vol. 50, no. 1, pp. 47–56, 2023

[O01] Y. Kim, H. Jeon, K. Choe, H. Song, B. Kim, and J. Seo, "Interactive visualization for exploring information fragments in software repositories," in *14th IEEE Pacific Visualization Symposium (PacificVis)*, 2021. (Poster)

Experiences

Research Experiences

Visiting Ph.D. Student | ***VIDi Lab. (advised by Dr. Kwan-Liu Ma)*** Jul.2022 – Dec.2022
University of California, Davis Davis, CA

Undergraduate Intern | ***Human-Computer Interaction Lab.*** Jan.2020 – Aug.2020
Seoul National University Seoul, Republic of Korea

Research Student | ***Algorithms Lab.*** Jun.2019 – Aug.2019
POSTECH Pohang, Republic of Korea

Research Student | ***Computer Vision Lab.*** Jan.2019 – Feb.2019
POSTECH Pohang, Republic of Korea

Teaching Experiences

Teaching Assistant

- Information Visualization and Visual Analytics, Seoul National University 2021 Fall
- Digital Computer Concept and Practice, Seoul National University 2020 Fall
  - Python programming
- Samsung DS^2 Program, Seoul National University 2021 Fall, 2021 Spring, 2020 Fall
  - Data visualization

Tutoring & Mentoring

- Undergraduate Research Opportunities Program, Seoul National University 2023 Spring, 2021 Fall
- CSE Tutoring Program, POSTECH 2020 Spring, 2019 Fall
  - Algorithms (CSED331)
  - Automata and Formal Languages (CSED341)

Academic Services

Reviewer

- IEEE Visualization and Visual Analytics (VIS) 2023, 2022
- Eurographics/IEEE VGTC Conference on Visualization (EuroVis) 2023, 2022
- IEEE Pacific Visualization Symposium (PacificVis) 2023, 2022
- ACM CHI Conference on Human Factors in Computing Systems (CHI) 2023, 2022
- ACM Conference on Intelligent User Interfaces (IUI) 2023
- Visual Informatics 2022, 2021
- Conference on Neural Information Processing System (NeurIPS) 2022

Student Volunteer

- IEEE Pacific Visualization Symposium (PacificVis) 2023
- IEEE Visualization and Visual Analytics (VIS) 2022
- ACM CHI Conference on Human Factors in Computing Systems (CHI) 2022

Awards & Honors

Outstanding Research Individual Fellowship | ***Brain Korea 21 (BK21)*** Mar. 2023
Granted 10,000,000 KRW (equivalent to 9,000 USD) Seoul, Republic of Korea

Star Student Researcher Award | ***Brain Korea 21 (BK21)*** Mar. 2023
Granted 700,000 KRW (equivalent to 600 USD) Seoul, Republic of Korea

Best Visualization Notes | ***15th IEEE Pacific Visualization Symposium (PacificVis 2022)*** Apr. 2022
Top 1st of all submissions [C02] Tsukuba, Japan

Jigok Scholarship | ***POSTECH*** Mar.2016 – Aug.2020
Full scholarship throughout undergraduate Pohang, Republic of Korea

Invited Talks

Measuring and explaining the inter-cluster reliability of multidimensional projections

- Korea Software Congress 2022 Dec. 2022

Toward a more reliable visual cluster analysis of high-dimensional data

- University of North Carolina, Chapel Hill (Dept. of Computer Science) Nov. 2022
  - Hosted by Dr. Ghulam Jilani Quadri and Dr. Danielle Szafir

Personal Info

Skills

Languages: English (Fluent), Korean (First Language)
Programming: c/cplusplus, Python, Java, JavaScript, TypeScript
Visualizations: WebGL, D3.js
Parallel Computing: OpenMP, CUDA, Numba

References

Jinwook Seo | ***Professor, Seoul National University*** Graduate Advisor
jseo@snu.ac.kr

Young-Joo Suh | ***Professor, POSTECH*** Undergradudate Advisor
yjsuh@postech.ac.kr